

Informatica Developer Tool Projects

ETL Project: Master Sorter, Joiner, Filter, and Aggregator

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Project Scope & Objective

Core Objective

Learn to:

- Join customer data from **two flat files**
- Filter data by **state**
- Aggregate **customer counts per area code**
- Store results into a **relational database**

Transformations Used

Sort Transformation → Joiner Transformation → Filter Transformation → Aggregator Transformation

Data Sources

Source 2: full_customers.csv

Column	Type
customer_id	string(11)
feedback_text	string
feedback_category	string
sentiment	string
complaint_intensity	int

Column (kept)	Type
customer_id	string(11)
state	string(2)
account_length	string(10)
area_code	string(3)
international_plan	string(3)
voice_mail_plan	string(3)
number_vmail_messages	int(10)

Dropped columns: total_day_minutes, total_day_calls, total_day_charge, total_eve_minutes, total_eve_calls, total_eve_charge, total_night_minutes, total_night_calls, total_night_charge, total_intl_minutes, total_intl_calls, total_intl_charge, customer_service_calls, churn, split, has_international_plan, has_voice_mail_plan, total_minutes, total_calls, total_charges, avg_charge_per_minute, support_call_rate, high_service_calls, usage_intensity, customer_value_segment, rule_based_churn_risk_score, rule_based_churn_risk_level.

Business Requirements

- 1 Sort the data in both source tables.
- 2 Join customers from both tables using `customer_id`.
- 3 Filter records where `state = 'KS'`.
- 4 Count customers per `area_code` within Kansas (KS).
- 5 Save the aggregated counts into a relational database.

Extra Analytics (Optional)

- Total customers with `international_plan = 'yes'`
- Total customers without `voice_mail_plan`
- Average `number_vmail_messages`

Implementation Steps 1-6

- ❶ Create project folder: `Project_CustomersInformation`
- ❷ Create subfolder: `DirCustomersInformation`
- ❸ Import `SRC1_customer_feedback` as flat file data object
- ❹ Import `SRC2_full_customers` as flat file data object
- ❺ Create map: `m_CustomersInformation`
- ❻ Import both sources into the map as `read` operations

Data Type Conversions

Source	Column	Converted Type
Source 1	customer_id	string(11)
	feedback_text	string
	feedback_category	string
	sentiment	string
	complaint_intensity	int (ordinal categorical)
Source 2	customer_id	string(11)
	state	string(2)
	account_length	string(10)
	area_code	string(3)
	international_plan	string(3)
	voice_mail_plan	string(3)
	number_vmail_messages	int(10)

Sorter, Joiner, and Filter Configuration

Sorter Transformations

- SRT1_CT (from Source 1) – sort on `customer_id` ascending
- SRT2_CT (from Source 2) – sort on `customer_id` ascending

Joiner: JN_CT

- Join type: **Normal**
- Condition: `customer_id1 = customer_id`
- Advanced: `sorted_input = checked`

Filter: FIL_CT

Condition: `state = 'KS'`
(All ports passed as input)

Aggregator and Target Table

Aggregator: AGG_CT

- Input: all ports from filter transformation
- New output port: CNT_CUSTOMERS = COUNT(customer_id)
- **Group by:** area_code

Action: Create DB connection in Informatica Developer, insert CustAgg as write operation in the map, then run the map.

Project Artifacts & Outputs

Screenshots included:

- ❶ 1OriginalSource1.png / 1OriginalSource2.png
- ❷ 2Source1Before.png / 2Source2Before.png
- ❸ 3SRT1_CT & 5SRT2_CT.png
- ❹ 4JN_CT.png
- ❺ 5FIL_CT.png
- ❻ 6AGG_CT.png & 7AGG_CT_output.png
- ❼ 8ProjectStructure.png
- ❽ 9TGT_CUSTAGG.png
- ❾ 10FullMapStructure.png

Appendices:

- A_Appendix_dataTypeConversion.png
- B_Appendix_nameConversion_data_Transformatio

Final Output: Table CustAgg populated with customer counts per area code for state = 'KS'.

ETL Pipeline Successfully Implemented

- Sorted sources on `customer_id` for efficient joining
- Joined datasets using JN_CT (Normal join, sorted input)
- Filtered Kansas customers with FIL_CT
- Aggregated counts per `area_code` using AGG_CT
- Loaded final aggregates into relational database table CustAgg

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